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To cite this article before publication: Caitlyn Lyons *et al* 2026 *Environ. Res.: Ecology* in press <https://doi.org/10.1088/2752-664X/aea99d>

Manuscript version: Accepted Manuscript

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Permafrost supports post-fire recovery of black spruce dominance in subarctic boreal forests

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Word counts (excluding figure captions)

Abstract: 298

Introduction: 1309

Methods and materials: 2790

Results: 1283

Discussion: 1865

Key words: Permafrost thaw, wildfire, boreal forest, black spruce, forest composition

Abstract

Black spruce (*Picea mariana*) dominates the boreal forest of North America, provides important wildlife habitat, and contributes substantially to long-term carbon storage. Wildfire shifts forest composition immediately following fire, but under climate change an intensifying wildfire regime may hinder black spruce regeneration and drive persistent compositional change away from black spruce. Further, wildfire can lead to permafrost thaw with unknown implications for forest regeneration. We monitored post-fire regeneration in black spruce stands in the Northwest Territories, Canada in permafrost and permafrost-free areas and explored how permafrost thaw rate (increased active layer thickness) impacted tree composition and density.

We measured tree composition during three time periods: pre-fire, 2-4 years post-fire, and 9 years post-fire. In permafrost areas, pre- and post-fire tree composition demonstrated greater similarity due to smaller shifts away from black spruce compared to permafrost-free areas. When evaluating post-fire tree densities with increasing time after fire, we found that regeneration outcomes became apparent more slowly in permafrost areas compared to permafrost-free areas. For example, in permafrost areas black spruce relative densities and dominance (>50% relative density) increased with time after fire, driven by mortality of trembling aspen and jack pine seedlings rather than ongoing black spruce recruitment. In permafrost-free plots, post-fire black spruce relative densities were more stable over the same period. In permafrost areas permafrost thaw rate also influenced post-fire black spruce densities; areas with rapid and moderate thaw had higher relative densities than areas with stable permafrost depths, though this effect diminished over time. Our study shows regeneration outcomes in forests underlain by permafrost are determined over a longer regeneration period than in permafrost-free areas. While permafrost persists in this landscape, permafrost presence and thaw determines post-fire regeneration outcomes, but climate-driven permafrost loss may exacerbate forest compositional shifts away from black spruce towards trembling aspen and jack pine.

Introduction

The boreal forest is dominated by coniferous trees and in boreal North America black spruce (*Picea mariana* [Mill.] B.S.P.; *Pinaceae*) is one of the most widespread and abundant tree species (McLeod and Macdonald, 1997). Wildfire is the dominant disturbance across the boreal biome, and black spruce is adapted to this regime through semi-serotinous cones and aerial seedbanks that promote rapid post-fire regeneration and long-term self-replacement, a strategy that has maintained its dominance for the past ~6000-3000 years (Hoecker et al., 2020; Kelly et

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al., 2013). Black spruce is also capable of persisting in cold, poorly drained permafrost environments (Van Cleve et al., 1981). However, climate change is driving accelerated warming in the boreal region compared to the global average rate (Wang and Dillon, 2014) resulting in increased wildfire activity (Young et al., 2017) and permafrost thaw (Gibson et al., 2018; Holloway and Lewkowicz, 2020; Schuur et al., 2008). Boreal trees, including black spruce, are strongly influenced by intensifying wildfire activity and permafrost thaw, which affect establishment and growth conditions. However, the interactions between wildfire and permafrost and their potential influence on post-fire regeneration processes remain poorly understood.

Permafrost is ground that is perennially at or below 0°C for two or more years (French, 2007), and ~80% of the boreal biome falls within the permafrost region (Helbig et al., 2016). Compared to faster-growing conifers and deciduous trees, black spruce has a conservative strategy characterized by a shallow rooting system and adventitious roots, enabling efficient use of nutrient-poor, organic soils and allowing it to persist and dominate in permafrost soils (Drobyshev et al., 2013; Van Cleve et al., 1981). However, changes in permafrost conditions may threaten the persistence of black spruce as a dominant species in these environments. Notably, thermokarst (abrupt thaw of ice-rich permafrost) can trigger physical landscape disturbances such as landslides and, in peatlands, ground subsidence, creating wet, hypoxic soil conditions that are intolerable to many tree species, including black spruce (Baltzer et al., 2014; Patankar et al., 2015). This can cause plant compositional changes, including shifts in dominance from black spruce to larch (*Larix laricina* (Du Roi) K. Koch) (Alexander et al., 2018; Dearborn et al., 2021), black spruce mortality (Dearborn et al., 2021), and forest loss (Baltzer et al., 2014; Helbig et al., 2016; Osterkamp et al., 2000). Conversely, gradual permafrost thaw which leads to active (seasonally thawed) layer thickening can exacerbate tree water stress. The water table

perches atop the permafrost, meaning that as permafrost thaws, the water table is found deeper in the soil profile limiting surface moisture in the rhizosphere (Quinton and Baltzer, 2013). Due to their shallow rooting system, vertical permafrost thaw reduces black spruce water use, leading to drought stress (Patankar et al. 2015), reduced growth (Sniderhan and Baltzer, 2016), and increased mortality (Dearborn et al., 2021). Rates of permafrost thaw are projected to increase across the high latitude boreal region, driven not only by climate warming but also by increased wildfire frequency, extent, and severity (Holloway et al., 2020). Wildfire removes insulating vegetation and organic soil layers, reduces evapotranspiration, and alters surface albedo, thereby increasing soil temperatures, deepening the active layer, and accelerating permafrost thaw (Gibson et al., 2018; Jafarov et al., 2013; Nossov et al., 2013; O'Donnell et al., 2011).

Climate change is intensifying wildfire activity across the North American boreal forest, driving shifts in post-fire tree regeneration and forest community composition (Baltzer et al., 2021). High-severity fires can reduce black spruce self-replacement mainly by combusting the soil organic layer and exposing mineral soil seedbeds that favor faster-growing species such as trembling aspen (*Populus tremuloides* Michx.) or jack pine (*Pinus banksiana* Lamb.), but can also consume black spruce's aerial seedbanks (Day et al., 2023; Fortin et al., 2026). Short-interval fires cause fires to occur before black spruce reach reproductive maturity and replenish their seeds, also contributing to reduced self-replacement (Brown and Johnstone, 2012; Hayes and Buma, 2021). These changes have contributed to compositional shifts away from black spruce toward deciduous, mixedwood, or unforested states across boreal North America (Baltzer et al., 2021; Day et al., 2023; Johnstone et al., 2010; Walker et al., 2023). Black spruce has shown evidence of stronger self-replacement in areas with longer fire return intervals, higher pre-fire black spruce basal area, poorly drained soils, lower canopy combustion, limited mineral soil

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3 104 exposure, and thicker post-fire soil organic layers (Brown et al., 2015; Day et al., 2023; Greene
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5 105 et al., 2007). Permafrost soils are often poorly drained and associated with thick soil organic
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7 106 layers creating cool, nutrient-poor conditions. Black spruce tolerates these conditions better than
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10 107 other tree species, allowing it to recruit and grow where faster growing species are limited
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12 108 (Johnstone et al., 2010; Van Cleve et al., 1983). However, post-fire site conditions in permafrost
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15 109 areas may not remain static; permafrost thaw can continue for years to decades after fire,
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17 110 reaching maximum thaw depths 5–10 years post-fire (French, 2007; Holloway et al., 2020). It is
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19 111 unknown whether post-fire changes in permafrost conditions might alter establishment
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21 112 conditions following fire thereby resulting in lagged forest recovery trajectories in permafrost
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24 113 areas.

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26 114 After fire, permafrost soils with thick residual organic soils should support black spruce
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28 115 self-replacement while hindering regeneration or reducing growth of other tree species that are
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30 116 less stress tolerant. However, wildfire is known to initiate rapid permafrost thaw (Dieleman et
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33 117 al., 2022; Gibson et al., 2018; Jafarov et al., 2013; Zhang et al., 2015). Deeper thaw alters the
34
35 118 growing conditions for plants by increasing the available rooting volume, soil temperatures,
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37 119 nutrient availability (Keuper et al., 2017; Salmon et al., 2016), and soil moisture conditions
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40 120 (Zastruzny et al., 2023). Wildfire has been shown to rapidly increase active layer thickness
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42 121 (ALT), the seasonally thawed layer above permafrost, within the first few years after fire
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44 122 (Dieleman et al., 2022; Holloway et al., 2020). Boreal trees also typically establish within the
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46 123 first five first five years after fire; recruitment in this period shapes regeneration outcomes for at
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48 124 least the next two to three decades (Gutsell and Johnson, 2002; Johnstone et al., 2020, 2004;
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51 125 Seidl and Turner, 2022). As permafrost continues to thaw post-fire, it can progressively alter soil
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126 conditions in ways that may disrupt initial tree regeneration patterns and lead to species-specific
127 differences in survival, causing lags in regeneration outcomes after fire.

128 In this study, we aimed to understand the interactive effects of wildfire and permafrost on
129 post-fire recovery of black spruce forests. We conducted the study in the southern Taiga Plains
130 of the Northwest Territories, Canada. We evaluated pre- and post-fire tree composition in areas
131 with and without permafrost, and quantified post-fire regeneration of black spruce, trembling
132 aspen, and jack pine using measures of density, basal area, and basal diameter. We further
133 examined how permafrost presence and other site conditions known to influence post-fire
134 regeneration affect regeneration outcomes. In permafrost areas, we quantified post-fire thaw
135 rates to assess whether increases in active layer thickness influence regeneration outcomes.
136 Specifically, we addressed two questions: 1) Does the presence of surface permafrost
137 (permafrost within the top 1.5 m of the soil profile) affect post-fire tree regeneration
138 composition? and 2) Does the rate of post-fire permafrost thaw affect regeneration composition?
139 We hypothesized that black spruce, which performs better on permafrost soils than faster-
140 growing species, would maintain dominance (>50% relative stem density) following fire where
141 permafrost is present. In areas without permafrost, warmer, nutrient rich soils may allow other
142 species to recruit more successfully (Gewehr et al., 2014), potentially altering competitive
143 dynamics over time. We further hypothesized that permafrost thaw represents a secondary
144 disturbance that could drive changes in regeneration patterns with time after fire. Rapid
145 permafrost thaw that improves soil conditions (e.g., increased rooting depth, warmer soils, and
146 greater nutrient and moisture availability) may allow black spruce to be outcompeted by more
147 resource-acquisitive, deeply rooted species such as trembling aspen. This study provides new

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insight into how permafrost presence and thaw rates influence post-fire tree regeneration composition.

Methods

Study region and species

Our study took place in the high boreal and mid boreal Taiga Plains ecozone of the Northwest Territories, Canada (Ecosystems Classification Group, 2009). The southern Taiga Plains ecozone falls within the sporadic discontinuous permafrost region and is a mix of undulating glacial till and peatlands. The topography in this area is relatively flat with rolling hills and is characterized by open to closed forested areas consisting of black spruce, jack pine, trembling aspen, balsam poplar (*Populus balsamifera* L.), larch, and less commonly white spruce (*Picea glauca* Moench.), and paper birch (*Betula neoalaskana* Sarg. and *B. papyrifera* Marsh., which co-occur here). The 30-year (1991-2020) annual temperature and precipitation normals for Hay River (~ 135 km from our nearest plot and the closest climate station) are -2.2°C and 320.3 mm, respectively (Environment and Climate Change Canada and Government of Canada, n.d.).

Historically, the Northwest Territories has experienced stand replacing fires every 70-100 years (Larsen, 1997). Boreal trees have several evolutionary adaptations for fire. Most notably, serotinous jack pine and semi-serotinous black spruce cones are stimulated to release seeds by fire, whereas trembling aspen and paper birch use long-distance seed dispersal and/or belowground resprouting to propagate after fire (Greene et al., 2007; Johnstone and Chapin, 2006). Our study took place following the 2014 wildfires, an extreme fire year that burned 2.85 million ha of forest in the Northwest Territories, approximately eight times the annual average (Walker et al., 2018b). With ongoing climate change the frequency of extreme fire years have

171 increased across the Northwest Territories (Hanes et al., 2025). Only 9 years later, in 2023,
172 another extreme fire year occurred where 3.5 million ha burned (Jain et al., 2024).

173 *Study design*

174 We choose 34 plots from within a large pre-existing plot network of 219 plots established
175 in 2015 to monitor forest recovery after the 2014 fires (see Day et al., 2023; Walker et al., 2018
176 for detailed methodology). We selected plots with and without pre-fire permafrost (17 of each)
177 (Fig. 1). These plots, located in three different burn scars, spanned mesic and subhygric drainage
178 categories, intermediate (60-100 years) and mature (>100 years) stand age classes, and low
179 (<33% proportion of organic soil burned), moderate (33-66%) and high ($\geq 66\%$) burn severity
180 classes (Supplemental Table 1). We selected permafrost-free plots that were similar to the
181 permafrost plots in drainage class, soil organic layer thickness, burn severity, and stand age to
182 minimize confounding factors when comparing the two groups (Supplemental Table 1). Black
183 spruce was abundant in the pre-fire stands for all plots with a mean relative abundance of 86%,
184 dominant at 31 plots, and co-dominant at three plots—determined by identifying dead standing
185 trees post-fire—and dominant in all but three of the plots.

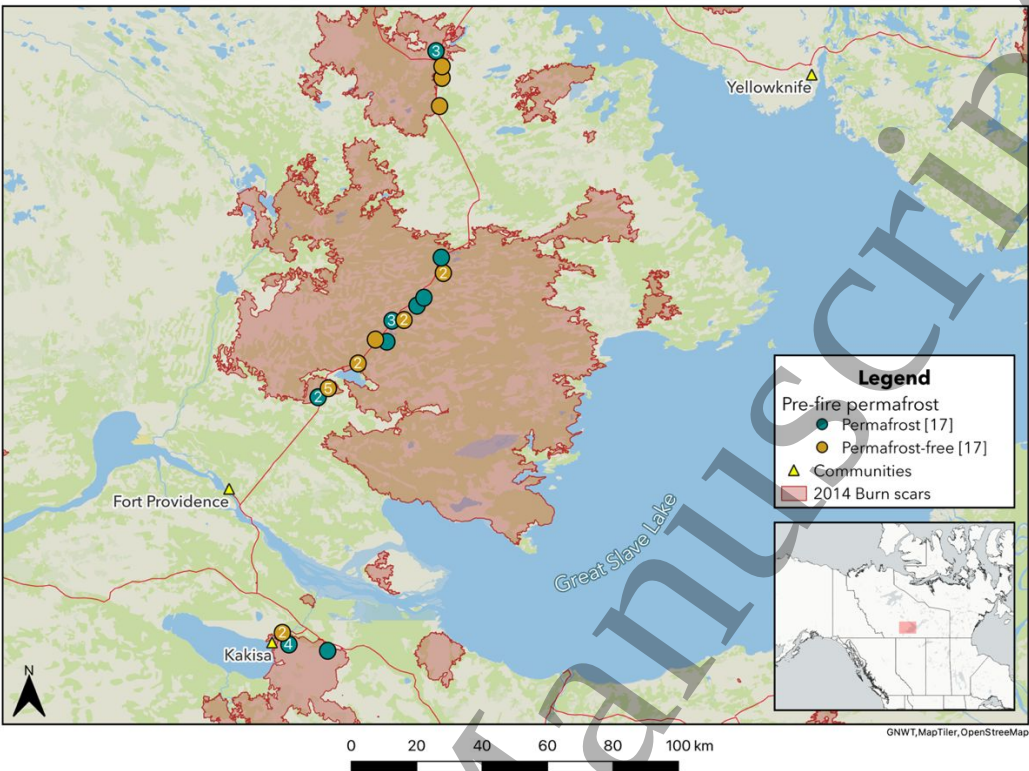


Figure 1. Map depicting locations of the 34, 60 m² plots within the southern portion of the Taiga Plains ecozone of the Northwest Territories, Canada established to quantify forest regeneration after the 2014 wildfires between plots with and without pre-fire permafrost. Where points locations are indistinguishable at the scale of the map, the number within the point indicates the number of plots found within that area. The inset map shows the location of the study region in western Canada.

At each plot we established a 30 m by 2 m (60 m²) belt transect. We determined stand compositional trajectories by identifying and counting juveniles within five 1 m² quadrats spaced evenly along the 30 m plot at two post-fire time periods: 2–4 years post-fire (2016–2018; each plot sampled once within this interval) and nine years post-fire (2023). 2-4 years post-fire has been shown to be a key recruitment window that can serve as an indicator of future stand composition (Charron and Greene, 2002; Johnstone et al., 2020, 2004; Seidl and Turner, 2022). Nine years post-fire provides a sufficient timeframe to document changes attributable to permafrost thaw as rapid permafrost thaw has been shown to occur within the first decade after

fire (Auclair et al., 2025; Dieleman et al., 2022; Holloway et al., 2020). Tree species identified included black spruce, jack pine, trembling aspen, and less commonly larch, balsam poplar, paper birch, and white spruce. We calculated both relative and absolute juvenile stem densities to represent post-fire tree regeneration. Stem counts were averaged across the five, 1 m² quadrats to create one absolute value of stem density (stems·m⁻²) per species per plot and time period after fire (i.e., 2-4 and 9 years after fire). Relative stem densities were calculated as stem density for each species relative to the total stem density in each plot. In 2023, nine years post-fire, we also estimated post-fire basal area and basal diameter (measured just above the root collar) for each tree species as larger recruits may be more likely to contribute to canopy dominance (Greene and Johnson, 1999). We measured basal diameters for three representative juveniles of each species present within each 1 m² quadrat. Individuals were selected to span the range of observed sizes within each quadrat, based on visual assessment of height and basal diameter. The mean of the three basal diameter measurements per species was used as the quadrat-level basal diameter and basal area (multiplied by the stem count). We then averaged these values across the five quadrats within each plot to obtain the plot-level average basal diameter (cm) and basal area (cm²·m⁻²) for each species (Table 1).

We assumed that if there was permafrost present at a plot the year after fire (2015), then there would have been permafrost pre-fire. The presence of permafrost pre-fire was determined in 2015 at five points evenly within the plot (every 6 m along the 30 m length of the plot), by inserting a 1.5 m steel rod vertically into the soil to measure the depth to resistance. If permafrost was detected at one or more points, the plot was deemed to have pre-fire surface permafrost. At plots with pre-fire permafrost, we measured maximum annual thaw depth (ALT) at the same five points from 2015 to 2023 in late September or early October, when seasonal thaw was complete.

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3 225 Changes in ALT through time were used to estimate post-fire changes in thaw depth. COVID-19
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9 227 In 2015 we measured other known drivers of post-fire forest composition including pre-
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11 228 fire basal area from mature trees (many/most were dead), exposed mineral soil, proportion of
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13 229 organic soil combusted, canopy combustion, stand age, drainage class, and residual soil organic
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15 230 layer thickness (Table 1; Johnstone et al. 2020). Pre-fire basal area describes pre-fire stand
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17 231 composition, percent exposed mineral soil cover indicates seedbed quality, stand age provides an
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19 232 estimate of time after last fire, and canopy combustion provides an estimate of fire severity. We
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21 233 determined pre-fire basal area ($\text{cm}^2 \text{ m}^{-2}$) and stand composition (% relative stem density) by
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23 234 counting, identifying, and measuring diameter at breast height (DBH; 1.4m) of every tree within
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25 235 each 60 m^2 plot. We estimated percent cover of exposed mineral soils for each quadrat within the
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27 236 plot and took the mean across all quadrats to determine a plot-level measure of percent exposed
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29 237 mineral soil. We estimated the proportion of organic soil combusted by comparing soil organic
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31 238 layer measurements after the 2014 fires to mature areas with no record of prior burning (see
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33 239 (Walker et al., 2018a, 2018b). We recorded canopy combustion for all trees within the 60 m^2 plot
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35 240 on a four-point scale 0-3: 0 – alive and no combustion; 1 – low combustion (only needles
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37 241 combusted); 2 – moderate combustion (many small branches remaining); 3– high combustion
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39 242 (only trunk and branch stubs remaining) (Walker et al., 2018). We used the mean canopy
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41 243 combustion across all trees within the plot as an estimate of plot-level canopy combustion
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43 244 severity. We estimated stand age at the time of fire (time after last fire) by collecting basal tree
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45 245 disks or cores and counting their rings. Disks and cores were sampled near the ground but above
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47 246 the root collar. Samples were taken from five individuals of each dominant conifer, either black
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49 247 spruce or jack pine and selected to represent the dominant size class in the stand thereby
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reflecting the most recent stand disturbance. Disks and cores were sanded until rings were clear and scanned. Ages were determined using Cybis CooRecorder v.7.8 (Larsson, 2006) or WinDendro 2009 (Regent Instruments Canada Inc. 2009). Ring counts demonstrated that tree ages were clustered (± 20 years) around recruitment cohorts and stand ages were determined based on the oldest tree in the recruitment cohorts (see Walker et al., 2018a). We assessed drainage class in the field based on topographic drainage, soil coarseness, and permafrost (Johnstone et al. 2008), which ranged from mesic with moderate surface moisture in flat upland terrain to subhygric with considerable surface moisture in lowland depressions. To characterize post-fire regeneration substrate conditions, we measured residual soil organic layer thickness adjacent to each quadrat within each plot by removing a small soil core (for more information see Walker et al., 2018a; Walker, Rogers, et al., 2018). See Table 1 for summary of measured variables.

Table 1. Summary and description of measured plot characteristics at 34 plots across the southern portion of the Taiga Plains ecozone of the Northwest Territories, Canada. These plots burned in 2014 wildfires.

Variable	Description	Mean ($\pm$ SE) or number of plots	Units
Pre-fire black spruce basal area	Pre-fire black spruce basal area	13.3 ± 1.46	$\text{cm}^2 \text{m}^{-2}$
Pre-fire trembling aspen basal area	Pre-fire trembling aspen basal area	0.001 ± 0.001	$\text{cm}^2 \text{m}^{-2}$
Pre-fire jack pine basal area	Pre-fire jack pine basal area	2.9 ± 0.81	$\text{cm}^2 \text{m}^{-2}$
Post-fire black spruce basal area	Juvenile black spruce basal area in 2023	0.64 ± 0.24	$\text{cm}^2 \text{m}^{-2}$
Post-fire trembling aspen basal area	Juvenile trembling aspen basal area in 2023	0.79 ± 0.15	$\text{cm}^2 \text{m}^{-2}$
Post-fire jack pine basal area	Juvenile jack pine basal area in 2023	0.79 ± 0.13	$\text{cm}^2 \text{m}^{-2}$
Post-fire black spruce basal diameter	Juvenile black spruce basal diameter in 2023	0.24 ± 0.02	cm
Post-fire trembling aspen basal diameter	Juvenile trembling aspen basal diameter in 2023	0.60 ± 0.10	cm

Post-fire jack pine basal diameter	Juvenile jack pine basal diameter in 2023	0.49 ± 0.13	cm
Pre-fire permafrost	Surface permafrost or permafrost-free	17 permafrost plots, 17 permafrost-free plots	unitless
Thaw rate category ¹	Stable (thaw rate -4 to 4 cm year ⁻¹), moderate thaw (>4 and ≤10 cm year ⁻¹), or rapid thaw (>10 cm year ⁻¹)	6 stable plots, 6 moderate plots, 4 rapid plots	unitless
Thaw depth	Mean maximum thaw depth for each thaw rate category	stable: 125 ± 6.80 moderate: 139 ± 5.99 rapid: 150 ± 0	cm
Exposed mineral soil cover ²	Mean percent cover of exposed mineral soil per m ² post-fire	3.55 ± 1.27	%
Drainage class	Mesic (moderate surface moisture and flat terrain) and subhygric (with considerable surface moisture and lowland depression). Based on Johnstone et al., 2008	16 mesic plots, 18 subhygric plots	unitless
Residual soil organic layer thickness	Post-fire soil organic layer thickness	25.6 ± 4.51	cm
Canopy combustion	Assessed on a four-point scale 0-3: 0 – alive and no combustion; 1 – low combustion (only needles combusted); 2 – moderate combustion (many small branches remaining); 3 – high combustion (only trunk and branch stubs remaining)	2.32 ± 0.11	unitless
Stand age	Stand age at the time of fire. Tree rings counted from basal discs or cores collected from dominant conifers present pre-fire	104 ± 6.39	years

¹Only 16 of the 17 pre-fire permafrost plots had enough annual measurements of thaw depth to calculate thaw rate.

²Although the plots in this study have thick soil organic layers there were still a small number of plots that had exposed mineral soil post-fire, specifically six plots had over 10% exposed mineral soil cover with a mean residual soil organic layer thickness of 11.5 cm.

Statistical analysis

All statistical analyses were performed using *R* version 4.4.2 (R Core Team, 2024) and all figures were created using “*ggplot2*” (Wickham, 2016) with “*ggeffects*” to extract model predictions (Lüdtke, 2018). Model assumptions were tested using the “*DHARMA*” package

(Harig, 2022) and visual inspection of residuals. Homogeneity of variances was assessed using Levene's test alongside visual inspection of residual boxplots. All model assumptions were satisfied.

1) Does the presence of surface permafrost affect the composition of post-fire tree regeneration?

We used two statistical approaches to assess whether pre-fire permafrost presence influenced post-fire tree composition. First, we related permafrost presence to changes in tree composition between pre- and post-fire conditions across two post-fire time periods (2–4 and 9 years). Second, we examined how permafrost presence influenced changes in post-fire juvenile densities (both relative and absolute) of black spruce, trembling aspen, and jack pine with increasing time after fire.

We assessed shifts between pre-fire and post-fire tree composition with principal coordinates analyses (PCoA) using the `{cmdscale}` function in the “*vegan*” package (Oksanen et al., 2024). We performed two PCoAs: the first to examine shifts between pre-fire composition and composition 2–4 years post-fire, the second was to examine shifts between pre-fire composition and composition 9 years post-fire. We specified a Bray–Curtis dissimilarity matrix computed with the `{vegdist}` function in the “*vegan*” package (Oksanen et al., 2024). We used relative stem densities for each tree species at each plot as the pre- and post-fire composition. We used arrows to indicate composition shifts between the pre- (tail of the arrow) and post-fire (head of the arrow) composition (Day et al., 2023). Longer arrows represent greater shifts away from pre-fire composition, and shorter arrows represent smaller shifts.

Axis 1 scores from the PCoA represented compositional variation away from black spruce and were related to permafrost presence and known post-fire regeneration drivers using

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3 295 linear regression in base R across all 34 plots. We quantified compositional change as the
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5 296 difference between pre-fire and post-fire Axis 1 scores, with larger values indicating greater
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7 297 shifts away from black spruce-dominated composition. We fitted two separate models to
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9 298 examine compositional change from pre-fire to 2–4 years post-fire and to 9 years post-fire.
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11 299 Predictor variables included permafrost condition (permafrost vs. permafrost-free), percent
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13 300 exposed mineral soil cover (seedbed quality), pre-fire black spruce basal area (seed availability),
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15 301 stand age (time after last fire), and canopy combustion (fire severity). The proportion of soil
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17 302 organic layer burned and post-fire soil organic layer thickness were excluded from models
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19 303 because both were associated with pre-fire permafrost presence (one-way ANOVA; proportion
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21 304 burned: $F_{1,66} = 7.84$, $p < 0.01$; soil organic layer thickness: $F_{1,66} = 21.68$, $p < 0.001$). We were
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23 305 most interested in the role of permafrost presence/absence in determining regeneration outcomes.
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25 306 Pairwise Pearson correlations were performed among all other predictor variables and were
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27 307 below $|r| = 0.5$. All continuous predictor variables were centered and standardized. To report
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29 308 model results we ran a type II ANOVA with the *car* package (Fox and Weisberg, 2019).
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35 309 The second analyses focused on post-fire forest regeneration of black spruce, trembling
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37 310 aspen, and jack pine, using relative and absolute stem densities of each species both 2-4 and 9
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39 311 years post-fire. Absolute density (stems m^{-2}) reflects stand-level recruitment, whereas relative
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41 312 density accounts for proportional changes in species composition independent of total stem
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43 313 abundance. We tested whether stem densities changed with time after fire, and the potential
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45 314 effects of permafrost condition using generalized linear mixed effects models (one model for
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47 315 each species, $n=34$). We modeled relative stem density using a binomial distribution, where
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49 316 species-specific stem counts were modeled relative to the total number of stems. We used a
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52 317 Poisson distribution for absolute stem density. All models were run using the “*glmmTMB*”
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package in *R* (Brooks et al., 2017) and we reported model results by running a type II ANOVA with the *car* package (Fox and Weisberg, 2019). As we used plot-level stem densities (juveniles m^{-2} averaged across all five quadrats), densities were rounded to the nearest whole number to meet the requirements for a binomial and Poisson model. Plots were repeatedly measured and therefore we included plot as a random effect. Predictor variables for all models included time after fire (2–4 vs. 9 years post-fire), pre-fire permafrost condition, percent exposed mineral soil, stand age, canopy combustion, and pre-fire basal area of the species under consideration. We also included the interaction between time after fire and pre-fire permafrost condition because we expected a lagged regeneration response in plots underlain by permafrost, as post-fire thaw alters site conditions over time. Post-fire organic layer thickness was excluded from all models because it was associated with pre-fire permafrost condition (see above for correlation results). Pairwise Pearson correlations were performed among all other predictor variables and were below $|r| = 0.5$. We centered and standardized predictor variables in all models. As plots were measured at two different time points, we included plot as a random intercept.

The tree species studied vary in terms of growth rates with trembling aspen and jack pine growth rates exceeding those of black spruce (Gutsell and Johnson, 2002); therefore, changes in relative stem densities may not represent an accurate depiction of future canopy dominance. Using the basal diameters measured 9 years post-fire we explored how basal area ($\text{cm}^2 \text{m}^{-2}$) and basal diameter (cm) varied across species and pre-fire permafrost condition 9 years post-fire. For both basal area and basal diameter, the residuals were not normally distributed, the data was heteroskedastic, and could not be corrected with transformation; therefore, we used Kruskal-Wallis tests to explore how basal area and average basal diameter varied across species and pre-fire permafrost condition.

2) Does the rate of permafrost thaw affect the composition of post-fire tree regeneration?

We explored the effects of permafrost thaw on post-fire relative densities of black spruce, trembling aspen, and jack pine, using post-fire changes in active layer thickness (ALT) to quantify permafrost thaw rates. We used annual maximum thaw depth measurements made between 2015 and 2023 at 16 of the 17 permafrost plots. One plot was omitted due to a lack of annual thaw depth measurements and permafrost-free plots were excluded. Permafrost thaw rates for each plot were calculated using the Siegel repeated medians method in the *R* package “*mblm*” (Komsta, 2019). We took the mean of all thaw depth measurements within each plot resulting in one value per plot per year because including a random effect increased model complexity and resulted in models that failed to converge. We placed plots in three different thaw rate categories: 1) stable permafrost – minimal post-fire change in ALT with flat regression slopes and thaw rates between -4 to 4 cm year⁻¹; 2) moderate thaw – moderate increases in post-fire ALT with positive regression slopes and thaw rates >4 and ≤10 cm year⁻¹; and 3) rapid thaw – steep, positive regression slopes indicating rapid post-fire increases in ALT, with thaw rates >10 cm year⁻¹ (supplemental Fig. 1). Because ALT > 1.5 m could not be measured, we truncated the data at 1.5 m and calculated thaw rates using only measurements collected prior to that threshold (supplemental Fig. 1). We used thaw rate categories to compare how different thaw scenarios influenced changes in the relative stem densities of black spruce, trembling aspen, and jack pine at 2–4 and 9 years post-fire

We ran one generalized linear model for each species in the “*glmmTMB*” package (Brooks et al., 2017) with the relative stem densities of each species as the response variable and repeated measures through time. We used binomial models and supplied the total stems for proportional response data and as such rounded plot-level relative stem densities to the nearest

whole number. Due to the low sample size for each model, we restricted our predictor variables to thaw rate category and measurement period after fire, with plot as a random effect. Rapid thaw plots were excluded from the jack pine model due to the absence of saplings 9 years post-fire. To ensure our thaw rate categories were appropriate, we also ran generalized linear models with thaw rate as the predictor variable. We used the same methods as above although the thaw rate category was replaced with thaw rate as a continuous predictor variable and the model output demonstrated similar results (supplemental Fig. 2 and Table 2).

Results

1) Does the presence of surface permafrost affect the composition of post-fire forest regeneration?

Although most study plots were dominated by black spruce at the time of fire in 2014, recruitment of jack pine and trembling aspen was common, in addition to the expected recruitment of black spruce. PCoAs representing pre- to post-fire compositional shifts indicated greater shifts away from black spruce in plots without permafrost compared to those with permafrost (Fig. 2). Both time periods of comparison showed a significant effect of permafrost presence on compositional shifts (pre-fire to 2-4 years post-fire: $F_{1,28}=4.74$, $p<0.05$, pre-fire to 9 years post-fire: $F_{1,28}=9.06$, $p<0.01$). However, the magnitude of pre- to post-fire compositional shifts decreased with time after fire in both permafrost and permafrost-free plots: model predictions of shifts away from black spruce were lower 9 years post-fire compared to 2-4 years post-fire (Fig. 2). None of the other measured predictors — pre-fire black spruce basal area, percent exposed mineral soil, canopy combustion, or stand age — significantly explained pre- to post-fire compositional shifts. This was true at both 2–4 years post-fire (basal area: $F_{1,28}=1.29$; mineral soil: $F_{1,28}=0.90$, combustion: $F_{1,28}=0.68$, age: $F_{1,28}=0.23$, all $p>0.05$) and 9 years post-fire

(basal area: $F_{1,28}=0.54$; mineral soil: $F_{1,28}=4.14$, combustion: $F_{1,28}=0.68$, age: $F_{1,28}=0.05$; all $p>0.05$).

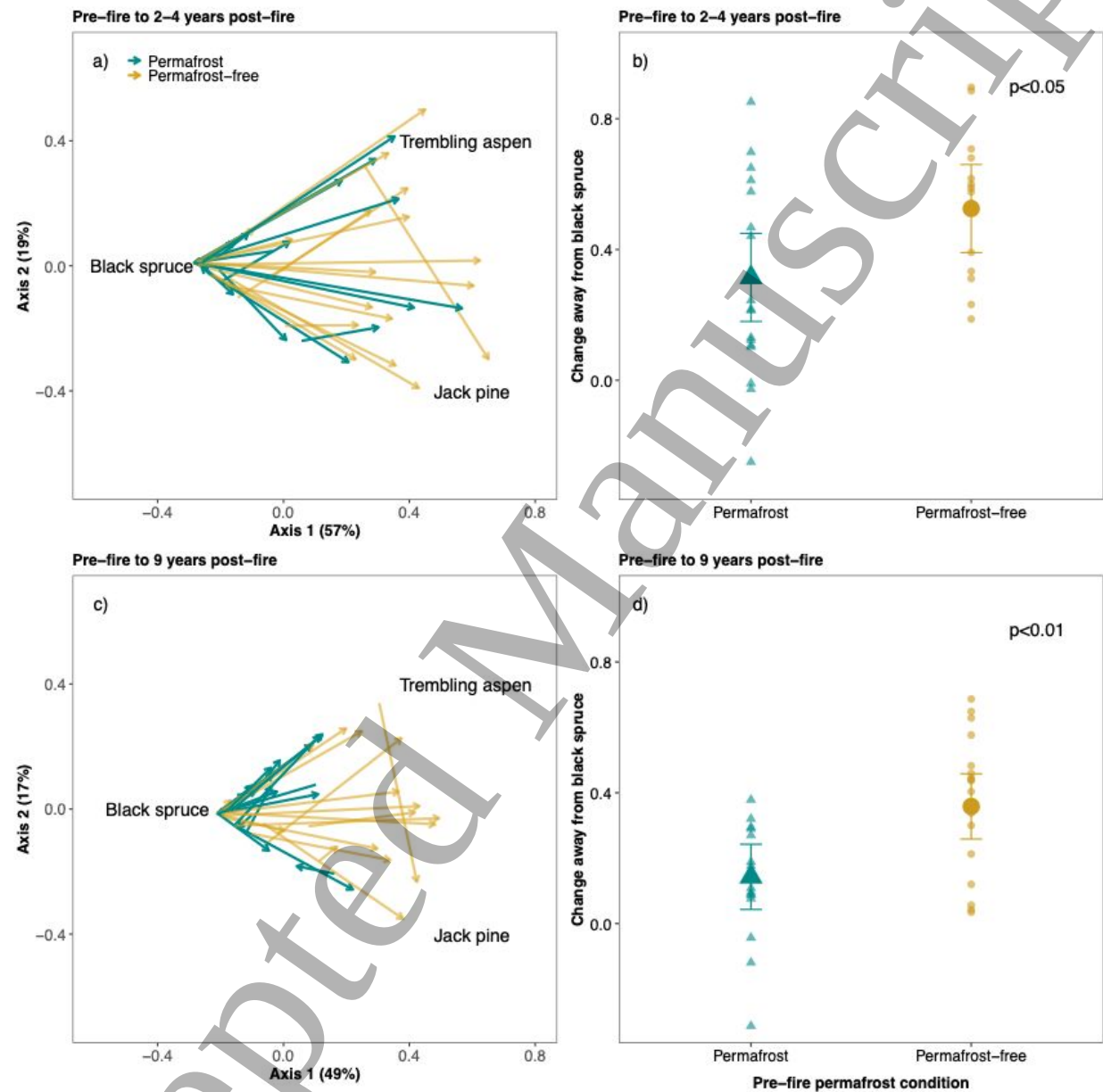


Figure 2. Greater pre- to post-fire canopy composition shifts away from black spruce in permafrost-free plots compared to permafrost plots. Principal coordinates analyses (PCoAs) illustrate changes in canopy species composition from pre-fire to 2–4 years (a) and 9 years (c) post-fire following the 2014 fires in the southern Taiga Plains, Northwest Territories, Canada ($n = 34$). Arrow bases represent pre-fire composition and arrow heads represent post-fire composition; longer arrows indicate greater shifts away from black spruce, while arrows pointing toward black spruce indicate shifts toward it in terms of relative abundance. Tree names shown on the plot mark approximate species centroids along the ordination axes (see Supplementary

Fig. 4). We used linear regressions to test whether pre- to post-fire compositional shifts differed between permafrost and permafrost-free plots. Shifts were calculated as differences in Axis 1 scores between pre- and post-fire communities in the PCoAs (a, c). Model predictions ($\pm 95\%$ CI) of compositional shifts away from black spruce are shown with raw data underlaid (b, d). Larger values denote greater shifts away from black spruce, and negative values denote shifts toward it. Significant p-values for permafrost effects are shown in the top right of each panel (b, d).

Black spruce recruitment after fire led to relative densities that were significantly higher in permafrost plots compared to permafrost-free plots ($\chi^2=11.13$, $df=1$, $p<0.001$) and these differences increased with time after fire ($\chi^2=36.20$, $df=1$, $p<0.001$) (Fig. 3a & 4a). Black spruce had the highest relative density in permafrost plots measured 9 years post-fire: specifically, 100% (17/17) of the permafrost plots were black spruce dominated ($>50\%$ of the stems m^{-2}) compared to only 53% (9/17) 2-4 years post-fire (Fig. 3a). Time after fire had a much smaller effect on measures of black spruce relative density in permafrost-free plots, with an increase in the number of spruce-dominated plots from 12% (2/17) measured 2-4 years post-fire to 29% (5/17) measured 9 years post-fire (Fig. 3a). With time after fire, black spruce relative densities recovered to values that were closer to pre-fire levels and this was most evident in permafrost-free plots; specifically in permafrost plots black spruce made up a mean of 87% of the stems pre-fire, reducing to 61% of stems 2-4 years post-fire, and increasing to 76% 9 years post-fire compared to permafrost-free plots that were 85% pre-fire, 32% 2-4 years post-fire, and 48% 9 years post-fire (Fig. 3 a,d).

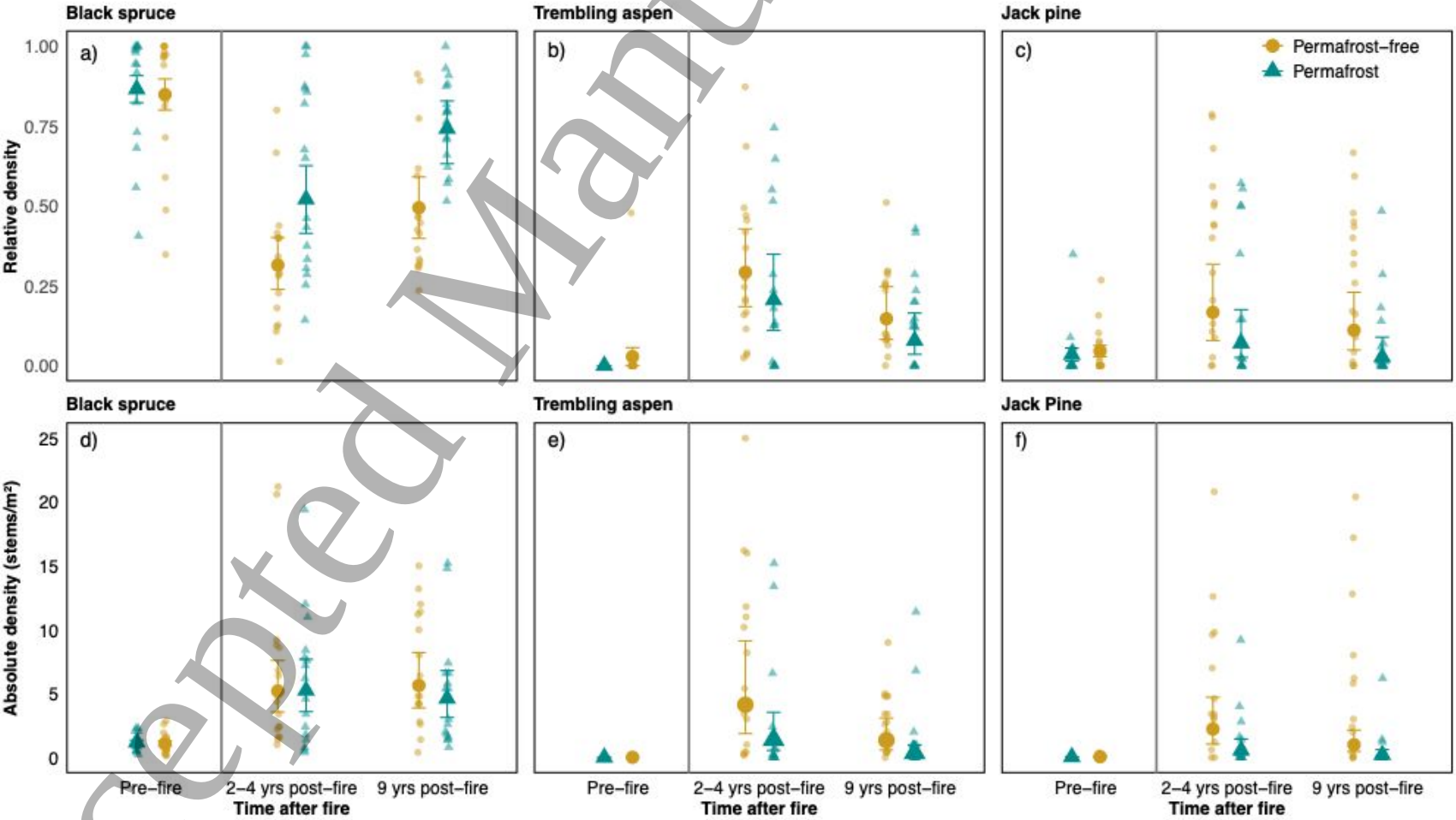
Relative densities of trembling aspen after fire did not differ between permafrost and permafrost-free plots ($\chi^2=1.24$, $df=1$, $p>0.05$) but showed an overall decrease with time after fire from 2-4 years to 9 years post-fire ($\chi^2=34.67$, $df=1$, $p<0.001$) (Fig. 3b & 4c). Similarly, jack pine relative densities did not differ between permafrost and permafrost-free plots ($\chi^2=3.57$, $df=1$, $p>0.05$) and also declined with time after fire ($\chi^2=9.57$, $df=1$, $p<0.01$). Jack pine seedlings showed the strongest decreases in relative density over time in permafrost plots: jack pine

relative density was twice as low in permafrost plots compared to permafrost-free plots when measured 2-4 years and was 3.5× lower by 9 years post-fire (Fig. 3c & 4e).

Higher black spruce relative densities in permafrost plots coincided with lower absolute stem densities of jack pine ($\chi^2=7.80$, $df=1$, $p<0.01$) and, although not significant, trembling aspen ($\chi^2=3.23$, $df=1$, $p>0.05$) while black spruce absolute densities were similar in both permafrost and permafrost-free plots ($\chi^2=0.13$, $df=1$, $p>0.05$) (Fig. 3, 4, & Supplemental Table 3). The increase in black spruce relative densities with time after fire was due to a decrease in absolute stem densities for both trembling aspen ($\chi^2=79.72$, $df=1$, $p<0.001$) and jack pine ($\chi^2=41.64$, $df=1$, $p<0.001$) with little change in black spruce absolute stem densities ($\chi^2=0.02$, $df=1$, $p>0.05$) (Fig. 3 c,d,e & Supplemental Table 3). Relative and absolute density changes between permafrost and permafrost-free plots were consistent with time after fire for all species and the interaction between time after fire and pre-fire permafrost was not-significant interaction ($p>0.05$) (Fig. 3 & 4).

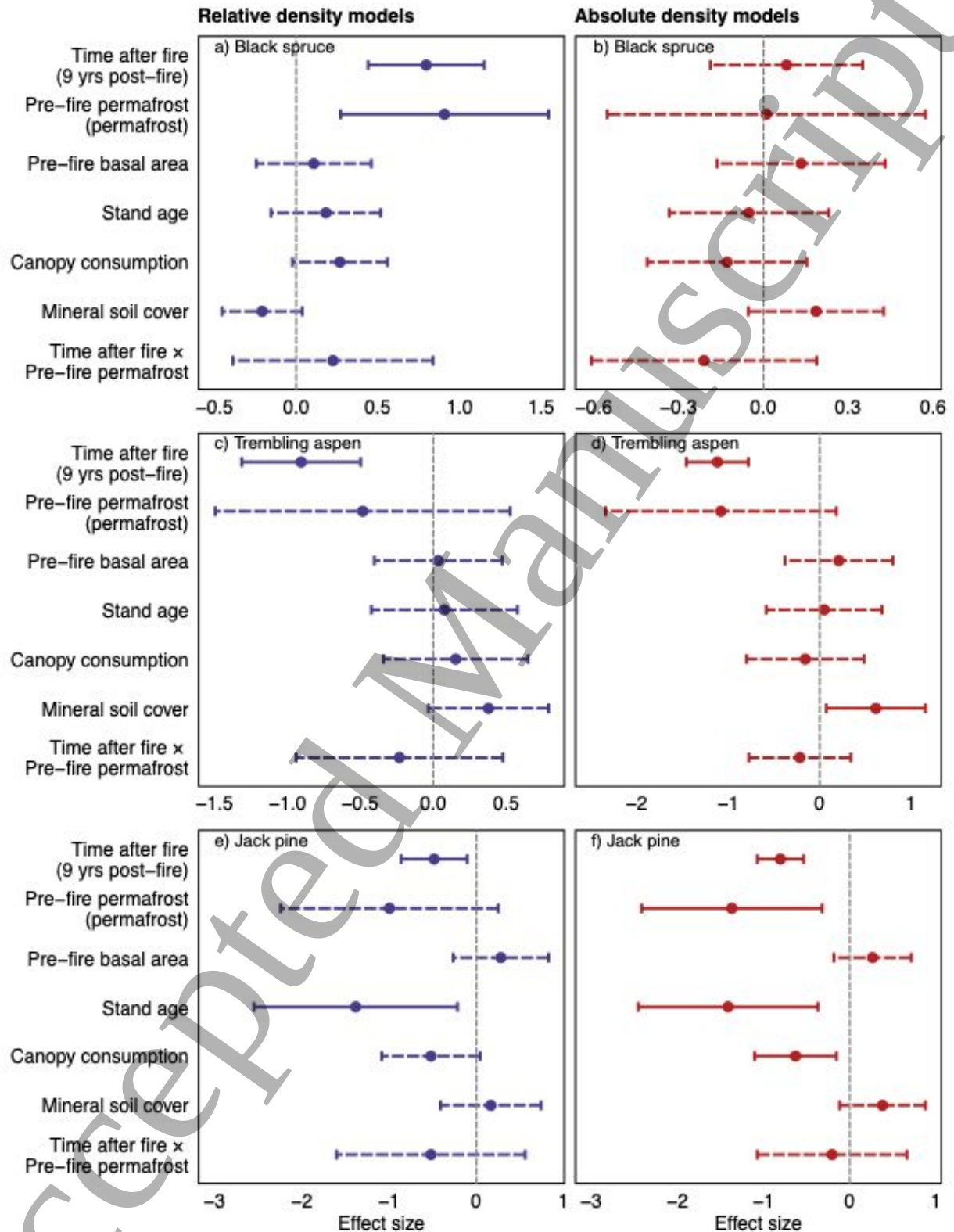
Most of the other abiotic and biotic factors (e.g., pre-fire basal area, percent mineral soil exposed, stand age and canopy combustion) were not significant predictors of post-fire relative densities of any species ($p>0.05$) (Fig. 4). However, jack pine relative densities decreased with increasing stand age ($\chi^2=5.64$, $df=1$, $p<0.05$) (Fig. 4). For absolute density there were more trembling aspen stems in plots where there was more exposed mineral soil ($\chi^2=5.16$, $df=1$, $p<0.05$) and there were less jack pine stems in plots with higher canopy combustion ($\chi^2=7.03$, $df=1$, $p<0.01$) and older stands ($\chi^2=7.35$, $df=1$, $p<0.01$); all other abiotic and biotic predictors were not significant ($p>0.05$) (Fig. 4).

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Figure 3. Black spruce relative density increases with time after fire and is higher in permafrost plots compared to permafrost-free plots, this is due to a decline in trembling aspen and jack pine absolute stem densities. Relative and absolute stem densities of black spruce (a,d), trembling aspen (b,e), and jack pine (c,f) regenerating after the 2014 fires in the southern portion of the Taiga Plains of the Northwest Territories, Canada. The x-axis denotes time after fire and symbols and colours denote either plots with or without pre-fire permafrost. Model predictions ($\pm 95\%$ CI) of relative/absolute stem densities are shown (larger point) with raw data underlaid as smaller points (N=34 plots). Pre-fire relative and absolute stem densities to demonstrate the comparability in pre-fire forest composition between permafrost and permafrost-free plots (see supplemental Table 3). Two absolute density values for both jack pine and trembling aspen were excluded from the plot because their high values obscured differences among group means. Separate generalized linear models were run for each species (binomial for relative density and Poisson for absolute density).



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463 **Figure 4. Post-fire relative and absolute stem densities are primarily driven by time after**
 464 **fire and pre-fire permafrost presence or absence. Estimated effect size results (± 2 standard**

errors) from generalized linear models exploring what factors influence post-fire relative and absolute stem densities regenerating after the 2014 fires in the southern portion of the Taiga Plains of the Northwest Territories, Canada. Separate models were run for each species: black spruce (a,b), trembling aspen (c,d), jack pine (e,f) using a binomial distribution for relative stem densities and Poisson distribution for absolute stem densities. The models explored how predictor variables (presented on the y-axis) influence relative and absolute stem densities for the respective species. Time after fire (2–4 and 9 years post-fire) and pre-fire permafrost (permafrost and permafrost-free) are both categorical variables with two levels; 2-4 years post-fire and permafrost-free are in the intercept. Effect sizes are presented as log-odds for relative density models and log-rate for Poisson models and as they are different models the effect sizes cannot be compared. Significant predictors ($p < 0.05$) are solid lines whereas non-significant predictors are dashed lines ($p > 0.05$).

Total basal area of juveniles trees measured 9 years post-fire did not differ among species ($\chi^2 = 0.99$, $df = 1$, $p > 0.05$) (supplemental Fig. 3a). This indicates that, although black spruce made up a greater relative density, trembling aspen and jack pine were larger in size (supplemental Fig. 3a). Indeed, the average basal diameters for trembling aspen ($0.6 \text{ cm} \pm 0.1 \text{ cm}$, mean $\pm$ standard error) and jack pine ($0.5 \text{ cm} \pm 0.1 \text{ cm}$) were more than twice that of black spruce ($0.2 \text{ cm} \pm 0.02 \text{ cm}$) (supplemental Fig. 3b). In addition, although basal area did not differ between species, it was lower overall in plots with permafrost ($0.5 \text{ cm}^2 \text{ m}^{-2} \pm 0.1 \text{ cm}^2 \text{ m}^{-2}$) compared to permafrost-free plots ($1.0 \text{ cm}^2 \text{ m}^{-2} \pm 0.3 \text{ cm}^2 \text{ m}^{-2}$) ($\chi^2 = 5.60$, $df = 1$, $p = 0.02$) (supplemental Fig. 3a). In contrast, the average basal diameters across all species did not differ between plots with and without permafrost ($\chi^2 = 1.07$, $df = 1$, $p > 0.05$) (supplemental Fig. 3b), indicating that the lower basal areas in permafrost plots can be explained by lower trembling aspen and jack pine absolute densities compared to plots without permafrost (supplemental Table 3).

2) *Does the rate of permafrost thaw affect the composition of post-fire forest regeneration?*

In the pre-fire permafrost plots, black spruce relative densities were lowest in permafrost-stable plots compared to moderate and rapid thaw plots (Fig. 5a & Table 2). This trend was most pronounced 2-4 years post-fire, where in both moderate and rapid thaw plots black spruce made

up 72% of stems compared to 41% in stable plots (Fig. 5a & Table 2). The lower black spruce relative stem densities measured 2-4 years post-fire in stable plots can be explained by higher trembling aspen (13.3 stems m^{-2}) and jack pine (2.53 stems m^{-2}) absolute stem densities in stable plots compared to moderate (trembling aspen: 2.93 and jack pine: 0.33 stems m^{-2}) and rapid (trembling aspen: 0.55 and jack pine: 0.45 stems m^{-2}) thaw plots (supplemental Table 4). With increasing time after fire black spruce relative densities became more similar across thaw rate categories, making up 71%, 80%, 75% of stems in stable, moderate, and rapid thaw, respectively, 9 years post-fire (Fig. 5a). Thaw rate category had no significant effect on trembling aspen and jack pine relative densities (Fig. 5b, c & Table 2). Our small sample size (16 plots) may have limited our ability to capture differences in trembling aspen and jack pine relative stem densities between thaw rate categories.

For both black spruce and trembling aspen there were significant interactions between time after fire and thaw rate categories (Table 2). Black spruce relative density increased with time after fire in stable permafrost plots but remained similar in moderate and rapid thaw plots (Fig. 5 and Table 2). Trembling aspen relative density decreased with time after fire in stable permafrost plots but there was small increase in rapid thaw plots (Fig. 5 and Table 2). All species' absolute stem densities decreased with time after fire across all thaw rate categories (black spruce and jack pine less so compared to trembling aspen) (supplemental Table 4). However, trembling aspen absolute stem densities saw a small increase with time after fire in rapid thaw plots (supplemental Table 4).

Table 2. Model parameters from generalized linear models exploring relationships between thaw rate category and time after fire against post-fire forest composition after the 2014 burns in the southern portion of the Taiga Plains of the Northwest Territories, Canada. Separate models were run for each species: black spruce, trembling aspen, jack pine. The response variables were the relative stem densities for each species, specified as a binomial proportional response. Predictor variables included thaw rate category with 3 levels: stable (-4 to 4 cm/year), moderate (>4 and

≤10 cm/year), and rapid (>10 cm/year) and time after fire with two levels: 2-4 and 9 years post-fire, and the interaction between thaw rate category and time after fire.

Black spruce				
<i>(black spruce relative density ~ thaw rate category × time after fire + (1 plot))</i>				
	estimate	std. error	z-value	p-value
Intercept	-0.73	0.29	-2.5	0.01
Thaw rate category [moderate]	1.60	0.45	3.53	<0.001
Thaw rate category [rapid]	1.92	0.64	3.01	0.003
Time after fire [9 years post-fire]	1.45	0.33	4.39	<0.001
Thaw rate category [moderate] * Time after fire [9 years post-fire]	-1.10	0.57	-1.93	0.05
Thaw rate category [rapid] * Time after fire [9 years post-fire]	-1.53	0.83	-1.86	0.06
Trembling aspen				
<i>(trembling aspen relative density ~ thaw rate category × time after fire + (1 plot))</i>				
	estimate	std. error	z-value	p-value
Intercept	-0.62	0.61	-1.02	0.31
Thaw rate category [moderate]	-1.08	0.88	-1.23	0.22
Thaw rate category [rapid]	-1.96	1.16	-1.69	0.09
Time after fire [9 years post-fire]	-1.6	0.38	-4.16	<0.001
Thaw rate category [moderate] * Time after fire [9 years post-fire]	0.95	0.64	1.48	0.14
Thaw rate category [rapid] * Time after fire [9 years post-fire]	2.18	1.11	1.97	0.05
Jack pine				
<i>(jack pine relative density ~ thaw rate category × time after fire + (1 plot))</i>				
	estimate	std. error	z-value	p-value
Intercept	-3.00	1.13	-2.66	<0.01
Thaw rate category [moderate]	-0.31	1.54	-0.20	0.84
Time after fire [9 years post-fire]	-0.40	0.60	-0.65	0.51

Thaw rate category
[moderate] * Time after fire -1.10 1.50 -0.74 0.46
[9 years post-fire]

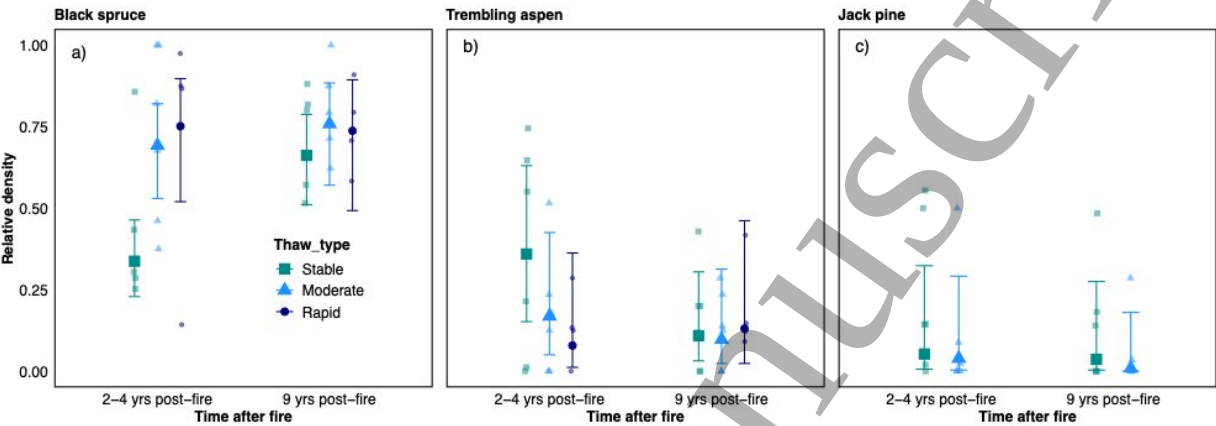


Figure 5. Black spruce relative density is lowest in stable thaw plots but increases with time after fire due to a decline in trembling aspen. Relative stem densities of black spruce (a), trembling aspen (b), and jack pine (c) at plots with different thaw rate categories based on post-fire changes in post-fire ALT (stable, moderate, rapid) regenerating after the 2014 fires in the southern portion of the Taiga Plains of the Northwest Territories, Canada. The x-axis denotes time after fire and symbols and colours denote different thaw rate categories (stable: -4 to 4 cm year^{-1} , moderate: >4 and ≤ 10 cm year^{-1} , rapid: >10 cm year^{-1}). Model predictions ($\pm 95\%$ CI) of relative/absolute stem densities are shown (larger points) with raw data underlaid as smaller points. Separate generalized linear models with a binomial distribution were run for each species. Rapid thaw is not included for jack pine models as there was no jack pine regenerating in rapid thaw plots 9 years post-fire.

Discussion

Permafrost presence and post-fire permafrost thaw were identified as key drivers of black spruce relative stem densities in burned black spruce forests in the Taiga Plains of Northwest Territories, Canada, with important consequences for post-fire tree composition. We selected permafrost and permafrost-free plots with comparable pre-fire characteristics (Supplemental Table 1), allowing us to confidently attribute the influence of permafrost on post-fire regeneration. We found that pre- to post-fire tree compositional shifts were greater when

measured 2-4 years post-fire than 9 years post-fire, particularly in permafrost areas where post-fire black spruce relative densities had rebounded by 9-years post-fire. Observed increases in black spruce relative densities were driven by trembling aspen and jack pine mortality with time after fire. Furthermore, as expected, we found evidence to support that rates of permafrost thaw can also influence post-fire regeneration. Although black spruce initially had lower relative densities in stable permafrost plots compared to moderate and rapid thaw plots, higher mortality of trembling aspen increased the relative dominance of spruce in stable permafrost with time after fire. Initial post-fire recruitment may demonstrate a mixed stand composition with greater densities of trembling aspen and jack pine, but mortality of trembling aspen and jack pine with time after fire in permafrost plots can allow black spruce to regain dominance. Previous studies show that when early post-fire conditions favor deciduous seedling establishment, deciduous dominance can persist for one to two decades post-fire (Johnstone et al., 2020, 2004). Our results suggest that in permafrost environments, early recruitment of deciduous species and fast-growing conifers may not persist, and post-fire species composition continues to shift during the first 9-years post-fire.

Our study provides evidence of post-fire compositional changes that are attributable to permafrost conditions, diverging from the typical post-fire recruitment trajectories. Jack pine has faster growth rates compared to black spruce (Gutsell and Johnson, 2002) and cold, nutrient-poor permafrost soils may not meet the resource requirements for this rapid growth that would be necessary to maintain post-fire establishment densities. Trembling aspen is also faster growing than black spruce (Gutsell and Johnson, 2002) and has been found to be most prevalent in areas with thicker active layers (> 120 cm) (Hopkins et al., 1955) and in warmer, more nutrient-rich, permafrost-free areas (Gewehr et al., 2014; Panda et al., 2010; Perala, 1990). However,

permafrost presence and thaw did not drive trembling aspen densities and in both permafrost and permafrost-free plots its densities declined similarly with time after fire. Jack pine absolute densities were almost 10x higher in permafrost-free plots compared to permafrost plots (supplemental Table 3); the permafrost-free soils supported initial jack pine recruitment but also lower mortality with time after fire. Our results suggest that both jack pine and trembling aspen can recruit post-fire both in plots with and without permafrost, but with time after fire trembling aspen and jack pine mortality led to increased black spruce relative densities, favouring black spruce in permafrost plots. We predicted that post-fire permafrost thaw would drive shifts in tree regeneration composition; however, we found that the presence of permafrost itself was a primary determinant of post-fire composition by maintaining higher black spruce relative densities and greater similarity to pre-fire stands.

In this landscape, permafrost is largely ecosystem-protected, meaning its persistence is strongly linked to local vegetation and soil conditions rather than climate alone (Shur and Jorgenson, 2007). Specifically, thicker organic layers insulate permafrost from incident radiation and permafrost is an impenetrable layer preventing drainage and leading to greater soil moisture, which promotes organic rich, nutrient poor, wet, cool soil conditions. Indeed, the permafrost plots had a deeper post-fire residual soil organic layer (38.5 ± 5.41 cm) compared to the permafrost-free plots (12.7 ± 1.14 cm). Recent literature suggests that black spruce is more resilient in areas with deeper residual soil organic layers that constrain the growth and establishment of other species including trembling aspen and jack pine (Baltzer et al., 2021; Day et al., 2023; Johnstone et al., 2020; Walker et al., 2023). In this study, because areas with deep soil organic layers were also underlain by permafrost, we could not isolate the effects of permafrost from residual soil organic layer thickness on post-fire recruitment.

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In addition to permafrost, we found that other site conditions known to influence post-fire regeneration were important predictors of post-fire trembling aspen and jack pine densities, but not that of black spruce. Specifically, exposed mineral soil was associated with greater trembling aspen recruitment, while younger stands with lower canopy combustion were associated with greater jack pine recruitment. However, these effects were not detected at the community level when comparing pre- and post-fire composition, likely because overall composition was more strongly driven by black spruce responses to time after fire and permafrost presence. The lack of an effect of pre-fire black spruce basal area, stand age, mineral soil cover, and canopy combustion suggests that permafrost presence may be an overriding driver of post-fire black spruce densities. Permafrost directly structures soil thermal and hydrologic conditions, creating cool, wet, nutrient-poor environments that may constrain post-fire establishment or survivorship of more resource demanding species. Our findings indicate that permafrost helps to reinforce patterns of post-fire black spruce self-replacement.

Previous studies have found that initial post-fire recruitment is a strong predictor for future stand composition (Gutsell & Johnson, 2002; Johnstone et al., 2020). Our findings suggest that permafrost and associated post-fire thaw can disrupt the initial post-fire forest regeneration trajectory following fire. Like Greene & Johnson (1999), who suggested that trembling aspen will start thinning almost immediately after fire and Johnstone et al., (2020), who reported thinning 10 years post-fire, we also observed increasing trembling aspen mortality with time after fire. Higher mortality in this case in permafrost plots led to a shift in tree seedling dominance towards black spruce. Trembling aspen’s reproductive strategy can influence the timing of post-fire thinning. Asexual reproduction via belowground sprouting promotes rapid early growth and can accelerate density-dependent thinning in the first-year post-fire, whereas

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3 614 sexual reproduction from seed results in slower initial establishment and delayed thinning.
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5 615 Because our study plots were conifer-dominated pre-fire and trembling aspen occurred in only
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7 616 one plot, post-fire regeneration was likely mostly from seed, supporting a more delayed thinning
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9 617 compared to when aspen reproduces asexually (Dawe et al., 2025; Day et al., 2020). Our results
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11 618 demonstrate that permafrost presence does not inhibit trembling aspen recruitment, but
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13 619 permafrost soils may not support establishment of trembling aspen past initial recruitment.
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15 620 Warmer, drier soils can retain trembling aspen up to at least 5 years post-fire (Dawe et al., 2022)
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17 621 and in our study up to at least 9 years post-fire in permafrost-free sites. Trembling aspen prefers
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19 622 warmer soils because cool soil temperature can reduce root growth suggesting that permafrost
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21 623 soils may limit trembling aspen growth potential (Wan et al., 1999). Other studies have noted
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23 624 thinning in lodgepole pine (a species closely related to jack pine) in the second (Johnstone et al.,
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25 625 2004) and third decade (Johnson and Fryer, 1989) after fire; we found jack pine mortality in the
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27 626 first decade after fire but initial recruitment was also lower in permafrost plots. This contrasts
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29 627 with black spruce that is both able to initially recruit and persist on permafrost soils for at least 9
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31 628 years post-fire, even in the face of thaw.

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33 629 In addition to permafrost impacting post-fire regeneration, wildfire-induced permafrost
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35 630 thaw is a secondary disturbance. Having annual thaw depth measurements post-fire allowed us
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37 631 to examine how permafrost thaw rate may be impacting post-fire regeneration outcomes. Our
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39 632 results demonstrate that permafrost thaw rate proved to be a significant predictor of post-fire
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41 633 black spruce relative density; this was most evident in stable permafrost plots where trembling
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43 634 aspen and jack pine mortality increased with time after fire allowing black spruce relative
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45 635 densities to increase. Previous studies that have looked at permafrost thaw impacts on trees in
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47 636 isolation from wildfire have found that with increasing thaw depths, shallow rooted black spruce
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3 637 can experience drought stress due to deepening water table that is perched atop the permafrost
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5 638 (Dearborn et al., 2021; Sniderhan & Baltzer, 2016; Patankar et al. 2015). This was also evident
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7 639 in our results as absolute black spruce stem densities was lowest in the rapid thaw plots that had
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9 640 the greatest thaw depths (>150 cm) compared to stable and moderate thaw (Table 2 for thaw
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11 641 depths and supplemental Table 4 for absolute densities). Stable thaw plots with their shallower
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13 642 thaw depths compared to rapid and moderate thaw plots may maintain access to water initially
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15 643 after fire and support the faster growing trembling aspen and jack pine.
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19 644 Previous studies at the landscape scale have found a decrease in forest cover with
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21 645 permafrost thaw especially in areas with thick organic layers and ice rich permafrost where
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23 646 subsidence has led to waterlogging and subsequent forest loss (Baltzer et al., 2014; Helbig et al.,
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25 647 2016; Osterkamp et al., 2000). In this study we measured changes in thaw depth (ALT) which
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27 648 precludes our ability to measure subsidence or lateral permafrost thaw as subsidence may limit
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29 649 changes in thaw depths. Wildfire can lead to subsidence due to permafrost thaw (Gibson et al.,
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31 650 2018; Holloway et al., 2020; Shur and Jorgenson, 2007) with potential impacts for post-fire
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33 651 regeneration. Future research should quantify how both vertical and lateral permafrost thaw
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35 652 impact post-fire regeneration as waterlogging due to subsidence can lead to poor seedbed
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37 653 conditions, reduced seedling survival and potential for regeneration failure (Camill et al., 2010).
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39 654 Post-fire thaw-induced subsidence is likely to increase black spruce mortality due to
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41 655 waterlogging, suggesting that that ice-rich permafrost may not support black spruce self-
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43 656 replacement (Dearborn et al., 2021). Due to the challenging nature of collecting annual thaw
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45 657 depth measurements our sample size in this study is low limiting our landscape level
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47 658 understanding of how permafrost thaw may be impacting post-fire regeneration.
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54 659 **Conclusion**
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Climate change is expected to lead to continued increases in both wildfire activity (Young et al., 2017) and permafrost thaw in boreal forests (Gibson et al., 2018; Holloway et al., 2020; Schuur et al., 2008). Our results support the idea that areas underlain with permafrost may delay ecosystem responses to climate change as they are maintaining permafrost out of sync with climate (Camill and Clark, 2000). In permafrost landscapes, black spruce showed greater resilience to fire compared to permafrost-free landscapes. The increase in black spruce relative densities was driven by trembling aspen and jack pine mortality with time after fire whereas black spruce absolute stem densities remained consistent, indicating that these landscape positions are serving as refugia for black spruce (sensu Stralberg et al. 2020). However, continued climate warming and wildfire activity will reduce permafrost prevalence across the north, reducing the availability of these refugia. In permafrost-free areas, enhanced recruitment and survival of jack pine and trembling aspen may create a persistently competitive environment, limiting black spruce self-replacement. Furthermore, where permafrost remains, continued vertical permafrost thaw beyond the biological zone of plants (i.e., rooting zone) may have implications for post-fire regeneration and these deep permafrost landscapes may no longer support post-fire black spruce dominance (Ogden et al., 2023). This study provides important first steps in understanding wildfire and permafrost thaw interactions on post-fire regeneration. We provide evidence of the role of permafrost in supporting maintenance of black spruce dominance across fire cycles in northern boreal forests.

Acknowledgments

We gratefully acknowledge the Wilfrid Laurier University — government of the NWT Partnership Agreement for providing logistical support and laboratory space. We also thank the many folks that contributed to field data collection and laboratory work. We also thank Brian

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Newton and Emma Sherwood for assistance in map production. We are grateful to the Aurora Research Institute, the Dehcho, Ka'a'gee Tu and Deh Gáh Got'ı́e First Nations, and the Tı́cho government for permission to conduct this work under research license 16815 and 17067.

Data availability statement

All data and R code will be freely available on Borealis.

Conflict of interest

The authors have no competing interests to declare

Funding statement

This article is Project 170 of the Government of the Northwest Territories Department of Environment and Natural Resources Cumulative Impacts Monitoring Program (funding awarded to JLB, JFJ, and Steve G. Cumming). Additional funding for this research was provided by NSERC (Changing Cold Regions Network), NSERC PDF to NJD, NSERC Discovery to MRT and JFJ, NSERC Canada Graduate Scholarship to CLL, Northern Scientific Training Program, NSF DEB RAPID grant no. 1542150 to MCM, NASA Arctic Boreal and Vulnerability Experiment (ABoVE) Legacy Carbon grant no. NNX15AT71A to MCM, and CFREF Global Water Futures — Northern Water Futures and Canada Research Chair funding to JLB.

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